Module 4 Practice Problems: Systems & Projections

Mathematical Foundations for Data Science & Computational Intelligence · Mohsin Raza · Fall 2026

Ungraded practice for Sessional 2. Everything here is solvable by hand: no code, no calculator needed. Problems marked ★ are the core set (about 90 minutes); do those first if you're short on time. Worked solutions are in a separate file. Try each problem before you look.
Notation. $A^\top$ is the transpose. $\hat{\mathbf{x}}$ is the least-squares solution. The residual is $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$.

Part A: When does $A\mathbf{x} = \mathbf{b}$ have a solution?

Problem 1 Solvable or not? ★

Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$.

(a) For $\mathbf{b}_1 = (2, 3, 4)$, find $\mathbf{x}$ with $A\mathbf{x} = \mathbf{b}_1$, or show none exists.

(b) Do the same for $\mathbf{b}_2 = (1, 3, 4)$.

(c) Explain your two answers using the *column picture*: what is special about $\mathbf{b}_1$ that $\mathbf{b}_2$ lacks? Then state each answer as a rank test, comparing $\text{rank}(A)$ with $\text{rank}([A \mid \mathbf{b}])$.

Problem 2 How many solutions?

For each matrix, say how many exact solutions $A\mathbf{x} = \mathbf{b}$ has for a *typical* right-hand side $\mathbf{b}$ (one chosen at random) (none, exactly one, or infinitely many). Then say whether the least-squares *coefficient vector* $\hat{\mathbf{x}}$ is unique.

(a) A is 3×3 with rank 3.

(b) A is 100×4 with rank 4 (100 data points, 4 parameters).

(c) A is 3×5 with rank 3.

(d) A is 4×3: its first two columns are independent, and its third column equals their sum.

Part B: Least squares and projections

Problem 3 A least-squares line by hand ★

Fit $t = x_1 + x_2 s$ to the four data points

$$(s, t) = (0, 1),\ (1, 2),\ (2, 4),\ (3, 5).$$

(a) Write down A and $\mathbf{b}$.

(b) Form the normal equations $A^\top A\,\hat{\mathbf{x}} = A^\top \mathbf{b}$ and solve them for $\hat{\mathbf{x}}$.

(c) Compute the fitted values and the residual vector $\mathbf{r}$. Check that $\mathbf{r}$ is perpendicular to *both* columns of A.

(d) Use the fitted line to predict t at $s = 4$.

Problem 4 Projection onto a single vector ★

Fit the model $\mathbf{b} \approx x\,\mathbf{a}$ (a single feature, no intercept), with $\mathbf{a} = (1, 2, 2)$ and $\mathbf{b} = (3, 0, 6)$.

(a) Find $\hat{x}$. (With one column, the normal equations reduce to a single fraction.)

(b) Find the projection $\mathbf{p} = \hat{x}\,\mathbf{a}$ and the residual $\mathbf{r} = \mathbf{b} - \mathbf{p}$. Verify that $\mathbf{r} \perp \mathbf{a}$.

(c) How far is $\mathbf{b}$ from the line through $\mathbf{a}$?

Problem 5 The projection matrix

Let $\mathbf{a} = (1, 1, 0)$. The matrix projecting onto the line through $\mathbf{a}$ is $P = \dfrac{\mathbf{a}\mathbf{a}^\top}{\mathbf{a}^\top\mathbf{a}}$.

(a) Write out P as a 3×3 matrix.

(b) Verify that $P^2 = P$ and $P^\top = P$. In one sentence each, say what these two properties mean geometrically.

(c) Compute $P\mathbf{b}$ and $(I - P)\mathbf{b}$ for $\mathbf{b} = (3, 1, 4)$. Check that the two results are perpendicular and add up to $\mathbf{b}$.

Problem 6 Reading residuals (short answers)

(a) In Problem 3, the residuals add up to exactly zero. Explain why, using the perpendicularity condition $A^\top\mathbf{r} = \mathbf{0}$.

(b) In Problem 4, do the residuals add up to zero? Compute their sum and explain the difference from (a).

(c) A colleague adds one more feature (one more column of A) to a regression. Can the training error $\|\mathbf{r}\|^2$ go *up*? Explain using column spaces.

Part C: Conditioning

Problem 7 A tiny change, a big difference ★

Consider $A = \begin{bmatrix} 1 & 1 \\ 1 & 1.001 \end{bmatrix}$.

(a) Solve $A\mathbf{x} = (2,\ 2.001)$.

(b) Now solve $A\mathbf{x} = (2,\ 2.002)$. The right-hand side changed by 0.001 in one entry. How much did the solution change?

(c) Explain what it is about the columns of A that makes the solution so sensitive.

(d) A's condition number is about 4×10^3. Up to how many significant digits could you lose solving systems with this A on a computer?

Problem 8 Condition numbers and the normal equations ★

(a) A diagonal matrix has singular values equal to the absolute values of its diagonal entries. Find $\kappa(A) = \sigma_{\max}/\sigma_{\min}$ for $A = \mathrm{diag}(20,\ 0.5,\ -0.002)$.

(b) A design matrix has $\kappa(A) = 10^4$. What is $\kappa(A^\top A)$? A computer stores about 16 significant digits. Up to how many digits could you lose with the normal equations, and up to how many with QR?

(c) Polynomial features $1, h, h^2, h^3$ are built from heights between 58 and 72 inches. Explain why A is badly conditioned, and name the standard fix.

Part D: Gram–Schmidt and QR

Problem 9 QR by hand ★

Use the same data as Problem 3: $\mathbf{a}_1 = (1, 1, 1, 1)$, $\mathbf{a}_2 = (0, 1, 2, 3)$, $\mathbf{b} = (1, 2, 4, 5)$.

(a) Normalize $\mathbf{a}_1$ to get $\mathbf{q}_1$.

(b) Subtract from $\mathbf{a}_2$ its projection onto $\mathbf{q}_1$. Which familiar data-preparation step is this? Normalize the result to get $\mathbf{q}_2$.

(c) Write down R (upper-triangular, with $A = QR$).

(d) Compute $Q^\top \mathbf{b}$, then solve $R\,\hat{\mathbf{x}} = Q^\top \mathbf{b}$ by back-substitution. Compare with your answer to Problem 3.

Problem 10 When the columns are already orthonormal

Suppose A has orthonormal columns $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ and $\mathbf{q}_2 = (0, 0, 1)$, and $\mathbf{b} = (3, 1, 4)$.

(a) Show that the normal equations reduce to $\hat{\mathbf{x}} = Q^\top \mathbf{b}$.

(b) Compute $\hat{\mathbf{x}}$ and the projection $Q\hat{\mathbf{x}}$.

Part E: Redundant columns and the pseudoinverse

Problem 11 Spot the redundancy ★

Each model includes an intercept (a column of ones). Which design matrices are rank-deficient? For each that is, give the exact linear relationship between columns.

(a) Price in Pakistani rupees and price in US dollars (converted at a fixed exchange rate).

(b) Temperature in °C and temperature in °F.

(c) A colour feature with three categories (red, green, blue), one-hot encoded as three 0/1 columns, all kept.

(d) A customer's age and their birth year (all ages measured in 2026).

For any one of the rank-deficient cases: are the model's *predictions* affected? Are its *coefficients* meaningful?

Problem 12 The pseudoinverse by hand

Let $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$ and $\mathbf{b} = (1, 3)$.

(a) Show that the normal equations can't be solved by inverting $A^\top A$.

(b) Since $A\mathbf{x} = (x_1 + x_2)(1, 2)$, only the sum $c = x_1 + x_2$ matters. Find the c that makes $\|c\,(1, 2) - \mathbf{b}\|$ smallest.

(c) Give three different vectors $\mathbf{x}$ that all achieve this best fit. Then, among *all* vectors $\mathbf{x}$ that achieve the best fit, find the one with the smallest length $\|\mathbf{x}\|$. (Hint: in the x_1x_2-plane, the best-fit vectors form a line.)

(d) Compute A^+ through the SVD: find the eigenvalues and eigenvectors of $A^\top A$, set $\sigma_i = \sqrt{\lambda_i}$, compute $\mathbf{u}_1 = A\mathbf{v}_1/\sigma_1$, and form $A^+ = \frac{1}{\sigma_1}\mathbf{v}_1\mathbf{u}_1^\top$. Check that $A^+\mathbf{b}$ matches your answer to (c).

Reading: Boyd & Vandenberghe, *VMLS* §§5.4, 8.3, 10.4, 11.5, ch. 12; Deisenroth et al., *MML* §§2.3, 3.8. Lecture slides: Module 4, Sessions 1–2.